

Business Organization

Business organization is a set up to perform specific functions to generate profit and in return provide service.

Business organization is an entity formed for the purpose to run a commercial enterprise. Business organization functions under relevant laws of the land.

Types of Business Organizations

There are 4 types of Business Organizations that function under the Companies Act 2017 duly registered with statutory body like Civil Engineering firm has to get itself registered with PEC

Entrepreneur

Entrepreneur is a person who takes risk to start – up business to earn profit and in return provide service

Entrepreneurship

Entrepreneurship is an activity of setting up a business or businesses, taking on financial risks

Entrepreneurship is to undertake business. It is a process to start – up business to plan, design, launch and manage and run business.

Business is often small but with hope to grow in future.

New business opportunities have encouraged entrepreneurship on a bigger or grand scale

Types of Entrepreneurships

Small business

Small business is defined as a privately owned firm or sole proprietorship that has fewer employees and less annual income than a scalable firm or regular-sized business.

Scalable Medium Business

Scalable business depends upon the direction and approach adopted by the firm. The firms' capability to grow without being adversely affected by available resources when faced with increased demand

Large Business Companies

Generally, large businesses are those in manufacturing like industries that employ 500 or more individuals

Social entrepreneurs

Social Entrepreneurs are those persons who render social service or community service

It is an emerging field and is growing rapidly and attracting increased attention from many sectors.

Social entrepreneurship is referred to entrepreneurs providing social service like in the social sector fields like establishment of education institutions or hospitals.

On the most basic level, there's something interesting and appealing about social entrepreneurs about why and how they do?

People are attracted to social entrepreneurs like:

1. Nobel Peace Prize laureate Muhammad Yunus for Grameen Bank in Bangladesh for micro financing
2. Aligarh Muslim University Old Boys Association AMUOBA who established SSUET,
3. Shaukat Khanum Memorial Hospital (SKMH)

These are examples in our country.

Extraordinary people came up with innovative ideas for creation a community service to transform life of communities for better future

But, interest in social entrepreneurship crosses boundaries of popularity and fascination

Social entrepreneurship signals the important reason to drive social change in society for betterment of society.

Types of Firms or Companies

Sole Proprietorship

Sole Proprietorship is the simplest form of business where a single person establishes a firm under It simply refers to a person who owns the business and is personally responsible for its debts

Sole Proprietorship is that form of business organization with only one owner of the business. The person performs 2 functions at a time a) organizer and business man b) Performer a manager, a supervisor or a risk-taker.

Merits

- 1) Easy to start
- 2) Easy to close
- 3) Quick decision
- 4) Need of small capital
- 5) Close contact with the customers
- 6) Business secrets remain intact
- 7) Free from labor disputes

Demerits

- 1) No scope of division of labor
- 2) No scope to get advantages of large-scale business
- 3) Possibility of high cost of production
- 4) Limited scope of using machines
- 5) Less credit worthiness
- 6) Uncertainty of continuity
- 7) Possibility to face economic crisis is very limited

Partnership

Partnership is a formal arrangement by two or more persons jointly own a business concern and to manage and operate a business and share its profits and losses to the extent of their shares or agreed proportion.

Merits or advantage

- 1) Easy to set up
- 2) Can secure more capital funds
- 3) Large-production is possible
- 4) Use of diverse talents
- 5) Promptness in decision – making
- 6) Check on hasty decisions

- 7) Personal interest and initiative
- 8) Availability of larger finances

Demerits or disadvantage

- 1) Lack of permanency
- 2) Lack of mutual confidence
- 3) Lack of liability
- 4) Delay in decision
- 5) Difficult to close down

Private Limited Company

Private limited Company is a privately held small business entity. A private limited firm can have 2 to 50 shareholders. The liability is limited to shareholders to the extent of their shares.. The liability of the members is limited to the extent of ownership of shares. It cannot raise funds from the market or stock exchange by way of trading shares

Public Limited Company (PLC) or Joint Stock Company

A public limited company (PLC) is the legal designation of a limited liability company that has offered shares to the general public and has limited liability. A PLC's stock is offered to the general public and can be acquired by anyone, either privately, during an initial public offering or through trades on the stock market. A public limited company can have any number of shareholders with liability of members limited. Public Limited company is listed in Stock Market like KESC, PTCL. It can raise funds from the market by way of shares and bonds.

It cannot start functioning unless minimum prescribed amount is available. It has to submit its annual report and balance sheet to the Registrar of Companies

- 1) Joint stock company is also known as the corporate form of business.
- 2) It is an association of individuals called shareholders who are authorized by the government to run a particular business.
- 3) Large-scale commercial and industrial enterprises are generally run under joint stock company system.
- 4) A joint stock company is formed according to the laws of the country.

- 5) Joint stock company can be set up under Companies Law.
- 6) There should be at least 7 persons, known as promoters, who would prepare plan of the company.

Merits

1. Can mobilize large resources
2. More Permanency
3. Specialized Management
4. Easy to collect capital
5. Transfer of shares
6. Suitable for larger enterprise

Demerits

- 1) Drawback of separation between ownership and management
- 2) Drawback of lack of interest of the shareholders
- 3) Exploitation of the majority of the shareholders
- 4) Drawback of limited liability
- 5) Burden of heavy taxes
- 6) Encourages Speculation
- 7) Disadvantages of large-scale production
- 8) Possibility of corruption
- 9) Possibility of Labor Disputes

Basic Investment Philosophy and Value Analysis:

The Basic Investment Philosophy and Value Analysis requires that before investment it is always desirable to conduct technical, financial and economic feasibility of project

There are various methods to determine financial economic feasibility through various mathematical models that serve as indicators
One of the method being Benefit Cost Ratio or B/C ratio

In Engineering Economics we will do a number of methods that help in preparing feasibility report of civil engineering or other projects

The Benefit Cost Ratio Method

Benefit Cost Ratio B/C or BCR or also called Benefit Cost Analysis (BCA). It is a technique that is widely used to assess or evaluate an alternate or a project to find out whether benefits are higher than costs or vice - versa. B/C or BCR or BCA is method to find out PV, annual cost and ROR. B/C provides a means for comparing benefits to costs.

The objective of Chapter on Benefit Cost Ratio is how to compare 2 alternatives on the basis of benefit/cost ratio. This method is required as a technique.

The higher the benefits the better

$$B/C = AWB/AWC$$

$B/C > 1$ Project desirable.

Methods of determining B/C

1. Conventional Method
2. Modified Method

Conventional Method

Conventional Method is older than Modified Method

In conventional method there is it is not possible to calculate Dis-benefit (DB)

As a first step calculate value of CR and then find value of B/C ratio

$$B/C = B / CR + O\&M \text{ ----- (1)}$$

$$CR = (I - F) [i(1+i)^N / (1+i)^N - 1] + F(i) \text{ ----- (2)}$$

Where

B = Annual Worth Benefit {AWB}

CR = Capital Recovery Cost – Initial investment converted to annual cost.

O&M = Operation & Maintenance Expenses of proposed project

F = Future Worth or Salvage Value (SV)

$B/C > 1$ Project Desirable.

Modified Method

Modified Method was derived later

In Modified Method there is provision to calculate DB

Equation:

$$B/C = B - DB - O\&M / CR \text{ -----(3)}$$

If DB is zero then equation would be

$$B/C = B - (O\&M) \text{ -----(3a)}$$

$$CR = (I - F) [i(1+i)^N / (1+i)^N - 1] + F(i) \text{ ----- (4)}$$

Where

B = Annual Worth Benefit (AWB)

CR = Capital Recovery Cost

O&M = Operation & Maintenance Costs

DB = Annual Dis-benefit

If there is no Disbenefit (DB) then equation will change:

$$CR = (I - F) [i(1+i)^N / (1+i)^N - 1] + F(i) \text{ -----(4)}$$

F = Future Worth

$B/C > 1$ Project is Desirable

Problem 1

Determine the B/C ratio by Conventional Method for a project with the following details:

	<u>Alternate A</u>	<u>Alternate B</u>
Initial cost	\$ 20,000	\$ 24,000
Project life	5 years	5 years
SV	\$ 4,000	\$ 4,000
Annual benefits	\$ 10,000	\$ 10,000
Annual O&M	\$ 4,400	\$ 4,400
Interest rate	8%	8%

Solution:

Given:

Initial Cost I	\$ 20,000
Project Life N	5 years
Future Worth For SV	\$ 4,000
Annual Worth Benefit B	\$ 10,000
Annual O&M	\$ 4,400
Interest rate (i)	8%

By Conventional Method

Formula:

$$B/C = B / CR + O\&M \text{-----} \quad \text{Eq(1)}$$

$$CR = (I - F) [i(1+i)^N / (1+i)^N - 1] + F(i) \text{--- Eq (2)}$$

Substituting values in Eq (2):

$$\begin{aligned} & (20000 - 4000)[8\%(1 + 8\%)^5 / (1 + 0.8\%)^5 - 1] + 4000(8\%)\} \\ CR &= (20,000 - 4000)\{8\%(1+0.08)^5 / (1+0.08)^5 - 1\} + 4000(0.08) \\ CR &= 16000 [0.08(1.08)^5 / (1.08)^5 - 1] + 4000(0.08)\} \\ CR &= 16000 [0.08(1.469)/1.469 - 1] + 320 \\ CR &= (0.1175/0.469) + 320 \\ CR &= 16000 (0.25) + 320 \\ CR &= 4000 + 320 \\ CR &= 4320 \end{aligned}$$

Substituting values in Eq (1)

$$B/C \text{ ratio} = 10,000 / 4,327 + 4400$$

$$10,000/8,727$$

$$B/C \text{ ratio} = 1.146 > 1.0$$

Problem 1 [Contd.]

By Modified Method DB being zero

$$\text{Formula } B/C = B - (O\&M) / CR \text{----- Eq (3)}$$

Substituting values in Eq(3)

Substituting value of CR from Conventional Method above

$$B/C \text{ ratio} = 10,000 - 4,400 / 4,327$$

$$= 1.294 > 1.0$$

$B/C > 1.0$ by either method the project is desirable and recommended

Problem 2

- Describe Benefit and Disbenefits of construction a Dam projects.
- Determine the B/C ratio based on modified method of projects for following given values:

<u>Details:</u>	<u>Alternate A</u>	<u>Alternate B</u>
Initial investment	\$ 300,000	\$ 500,000
Annual Income	\$ 25,000	\$ 25,000
Annual O&M	\$ 5,000	\$ 5,000
Annual Dis-benefit	\$ 2,000	\$ 2,000
Project life	25 years	25 years
SV or F	\$ 0	\$ 0
Interest rate	10%	10%

Solution:

Given:

Initial investment	\$ 500,000
i	10%
N years	25 years
Annual Income (B)	\$ 25,000
DB	\$ 2,000
Salvage Value F	\$ 0

By Modified Method

Formula: $B/C = B - DB - (O\&M) / CR$

$$CR = (I - F) [i(1+i)^N / (1+i)^N - 1] + F(i)$$

Substituting values in above equations

B/C

Project B Contd.

$$CR = (I - F) \left[\frac{i\%(1 + 10\%)^n}{(1 + i\%)^n - 1} \right] + SV (i\%)$$

$$CR = (500,000 - 0) \left[\frac{10\%(1 + 10\%)^{25}}{(1 + 10\%)^{25} - 1} \right] + 0 (10\%)$$

$$CR = 500,000 \left[\frac{0.1 (1.1)^{25}}{(1.1)^{25} - 1} \right]$$

$$CR = 500,00 \left[\frac{0.1 (10.8)}{(10.8) - 1} \right]$$

$$CR = 500,000 \left[\frac{1.08}{9.8} \right]$$

$$CR = 500,000 (0.11)$$

$$CR = 55,000$$

$$B/C = \frac{B}{CR + O\&M}$$

$$B/C = \frac{25,000}{55,000 + 5000}$$

$$B/C = \frac{25,000}{55,000 - 5,000}$$

$$B/C = \frac{25,000}{60,000}$$

$$B/C = 0.41$$

Project being > 1 is not desirable

$$B/C = (25000 - 2000 - 5000) / (500,000 - 0) [0.1(1.1)^{25} / (1.1)^{25} - 1] + 0$$

$$B/C = 18,000 / 500,000[0.11]$$

$$B/C = 18,000 / 55,000$$

$$B/C = 0.327$$

0.327 < 1.0 Project not desirable